



AI in Robotics

Making way into our bodies?

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In the era of invasive smart technologies, both personal and professional lives of humans are undergoing a sea of changes. Thanks to electronic evolution that is impacting our daily chores from the bedroom to the boardroom as well as in public spaces. Be it voice assistants, chat bots, automated and gesture-driven devices, an immersive experience with VR (virtual reality) goggles or multi-media features, drones and robots, the influence of artificial intelligence (AI) and machine learning is hard to miss in every domain we come across. Some may argue against its aggressive intrusion, while others may call it a boon to mankind for its speed, accuracy and efficacy. From transportation to banking, retail, fashion, energy and utility sectors to even health, digitalization is steadily deepening its roots to rule the roost. Robotic surgery is one such break-

through in medical science that is conspicuously leaving its mark in OTs.

AI in OT rooms

Robot-assisted surgery is making waves in surgical fraternity for its precision, hassle-free procedures and improved outcomes. This method is applied with the help of small instruments attached to a sophisticated, flexible robotic arm. The surgeon regulates the mechanical arm via a computer. Such trailblazing surgeries allow doctors to execute complex operations with more perfection than what is



achieved through traditional practices. Commonly involving minimally invasive surgery performed through tiny incisions or small cuts, robotic surgeries are already successfully carried out on the eye, knee, kidney, gallbladder, in cancerous conditions, plus in laparoscopy and hernia operating procedures, in digestive system or gastrointestinal surgery, in colorectal surgery as well as in cases related to oncology, urology, orthopedic and cardiology, among others.

AI can uplift robotic surgery by mitigating the surgeons' stress. A slew of AI algorithms used in such innovative surgeries includes machine learning, computer vision and deep learning. This streamlines the heavy-duty surgical procedures by zeroing in on the tools, supervising operations and conveying warning signals. Thus, an AI-based system is capable of guiding surgical procedures with higher efficiency.

In robot-assisted surgery, AI and machine learning are used for motion planning, computer vision, force sensing, teleoperation and predictive analytics. These algorithms and models optimize the trajectory of the robotic arm, provide real-time feedback to the surgeon, adjust the force applied by the arm, enhance the user interface of the console and guide surgical decisions. "As technology evolves, we can expect to see even more advanced applications of AI and machine learning in robotic surgery," volunteers Dr. Jaideep Jesson Rayapudi, MBBS, MD, serving as a senior subject matter expert in Healthcare AI at iMerit where he helps create large-scale datasets to power a variety of healthcare AI machine-learning use cases.

Da Vinci leads the way

The Da Vinci robotic system employed in complex surgeries is manufactured by Intuitive Surgical, Inc., an American company that develops, manufactures and markets robotic products designed to upgrade clinical outcomes of patients through minimally invasive surgery, most notable being the Da Vinci Surgical System. It is used in prostatectomy, cardiac valve repair as well as for renal